

Karandeep Singh

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Science (Spec in AI), Minor in Philosophy

- GPA: 3.5/4.0
- Relevant Coursework: Data Structures&Algorithms, Foundations in ML, OOD, Discrete Structures, Linear Algebra

EXPERIENCE

Software Engineering Intern (Remote)

Jan 2025 – Present

ThinkGrid Labs

Kerala, India

- Developed and maintained frontend UI/UX features across multiple pages for Hyrelyst.com, handling user authentication and dashboard functionality
- Transitioned to full-stack development, building backend APIs and database integration using NodeJS and MongoDB, and managing a database of 300+ user records
- Collaborated with senior engineers using Git version control and agile development practices

RESEARCH

AI and Data Ethics Group (AIDE)

Jan 2026 – Present

Northeastern University

- Contributing member of research group focused on AI ethics, policy analysis, and responsible AI development
- Drafted and contributed language to proposed Massachusetts digital replica legislation; provided analytical suggestions, receiving positive feedback from group leads
- Engaged in ongoing analysis of AI policy proposals and ethical frameworks affecting emerging AI technologies

PROJECTS

Radeye | *Computer Vision, ML, PyTorch*

Apr 2026

- Built a real-time eye-tracking pipeline using MediaPipe Face Mesh and OpenCV to capture iris position and velocity from webcam at 30fps
- Collected 18 typing test sessions and extracted velocity features (avg, max, std deviation) for correlation analysis between eye movement and typing performance
- Found moderate positive correlation ($r=0.469$) between average eye velocity and WPM, with weak negative correlation ($r=-0.321$) for max velocity
- Trained a PyTorch feedforward neural network to predict WPM from eye movement features, achieving test RMSE of 36 WPM

Language Feedback API | *FastAPI, Anthropic Claude API, Python*

Mar 2026

- Migrated boilerplate from OpenAI to Anthropic Claude Sonnet, designing a single-call prompt strategy with temperature tuning to deliver consistent JSON-formatted feedback within a 30-second response window
- Engineered prompts iteratively against multilingual test cases (Japanese, Portuguese), refining rules to handle phrasal grammar, conversational word choice, and direct-translation pitfalls
- Implemented JSON response validation as a safety net against malformed LLM output, plus unit, schema, and integration test suites with pytest
- Containerized with Docker and Docker Compose for reproducible deployment

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript

Web Development: Tailwind CSS, MongoDB, NodeJS, NextJS, React, Framer Motion, Shadcn

ML/Data Science: Pandas, Matplotlib, NumPy, Scikit-learn, PyTorch, MediaPipe, OpenCV, FastAPI

Tools: VS Code, Git